Jetson Edge LLM Model Selection Guide

2B & 3B parameter LLM guide for NVIDIA Jetson 8GB. Covers 24 models: strengths, best workflows, benchmark speeds, and quick-pick recommendations. Compiled August 18, 2026.

Benchmark method: llama.cpp with -ngl 99 (GPU offload), Q4_K_M quantization, 200-token generation, flash attention. Models with unsupported GGUF formats benchmarked via Ollama API. Prompt: "Explain the difference between hypothyroidism and hyperthyroidism in 3 sentences."

Part 1: Verified Benchmark Results

Models tested and verified on this Jetson 8GB device. Speeds are generation tok/s unless noted.

Model	Params	Gen tok/s	Prompt tok/s	Method	Notes
gemma4:e2b	2B	27.1	41.5	llama.cpp	Fastest generator
gemma2:2b	2.6B	25.1	117.1	llama.cpp	Strong generalist
lfm2.5:2.6b	2.7B	25.1	115.0	llama.cpp	Best for agentic/tool use
qwen2.5:3b	3.1B	21.3	258.4	llama.cpp	Best math at 3B
phi3:3.8b	3.8B	21.2	186.9	llama.cpp	Strongest reasoning/param
hermes3:3b	3B	20.7	180.3	llama.cpp	Best function calling at 3B
llama3.2:3b	3.2B	20.6	293.7	llama.cpp	Best ecosystem support
qwen3.5:2b	2B	8.7	120.9	Ollama API	New RoPE unsupported by llama.cpp

Remaining models are still downloading and will be benchmarked as they arrive.

Part 2: Model Guide - All 24 Models

Organized by model family. Each entry includes developer, architecture, strengths, best workflows, and cautions.

Google Gemma Family

Model	Dev	Params	Key Strengths	Best Workflow	Cautions
Gemma 2 2B	Google	2.6B	Knowledge distillation training; punches above weight; strong writing quality	Local chat assistant, summarization, general text generation on edge	8K context only; text-only; struggles with complex reasoning
Gemma 3 1B	Google	1B	Extremely compact (~529MB int4); 32K context; QAT baked in; multilingual	On-device/mobile text processing, offline assistants on extreme low-memory hardware	1B params limit capability; no vision; best for simple text only
Gemma 3n E2B	Google	5B raw / 2B eff.	Multimodal: text+image+audio+video; MatFormer elastic inference; 140+ languages; 32K context	Edge multimodal AI: medical image analysis, audio transcription, video frame processing	~5B on-disk footprint; 10-15% accuracy loss vs E4B; complex deployment
CodeGemma 2B	Google	2.6B	Specialized FIM code completion; trained on 500B+ code tokens; IDE integration	IDE autocomplete / fill-in-the-middle; NOT a chat model	Not instruction-tuned; outputs nonsense as chat; older Gemma 1 arch; use 7B-IT for code chat

Alibaba Qwen Family & Microsoft Phi

Model	Dev	Params	Key Strengths	Best Workflow	Cautions
Qwen 2.5 3B Instruct	Alibaba	3.09B	Exceptional math (65.9 MATH, 86.7 GSM8K); 29+ languages; 32K context (128K extended); Apache 2.0	General-purpose instruction following, multilingual Q&A;, math problem solving, structured data extraction	World knowledge limited at 3B; instruction strictness below Phi-3.5
Qwen 2.5 Coder 3B	Alibaba	3.09B	Code-specialized on 5.5T code tokens; 92+ programming languages; same efficient 3B footprint	Code generation, completion, review/debugging, SQL generation, developer assistant	Narrower general NLP ability; not for open-ended knowledge Q&A; or creative writing
Qwen 3.5 2B	Alibaba	2B	Hybrid DeltaNet+Attention; dual thinking/non-thinking modes; multimodal (vision); 262K context; 201 languages	Minimal VRAM multimodal tasks, long-context documents, multilingual applications, thinking-mode reasoning	Thinking mode must be explicitly enabled; very new (Feb 2026); fewer community benchmarks; llama.cpp unsupported
Phi-3 Mini 3.8B	Microsoft	3.8B	Rivals Mixtral 8x7B on reasoning (69% MMLU); edge-optimized CPU-first design; 128K context variant	On-device reasoning, math/logic problems, structured extraction, RAG with search augmentation	Default 4K context; limited world knowledge (authors recommend search augmentation); English-focused

Meta Llama & Fine-tunes

Model	Dev	Params	Key Strengths	Best Workflow	Cautions
Llama 3.2 3B Instruct	Meta	3.21B	8 multilingual languages; 128K context; edge-optimized with native quantization; best ecosystem support	Reliable baseline assistant, multilingual edge inference, RAG with long context	No native tool calling; weak coding at 3B; knowledge cutoff Dec 2023; Llama license restrictions
Hermes 3 3B	Nous Research	3.21B	Full fine-tune of Llama 3.2 3B; advanced function calling + JSON structured output; strong roleplay; user-steerable	Agentic function-calling workflows, structured JSON output, roleplay/character assistants, multi-turn agents	Coding still weak at 3B; permissive alignment may need evaluation for clinical use; inherits Llama knowledge limits

Model	Dev	Params	Key Strengths	Best Workflow	Cautions
Ministral 3B	Mistral AI	~3B	Purpose-built edge tier; 128K-256K context; strong multilingual (European + Asian); efficient dense architecture	On-device multilingual assistants, long-context RAG, privacy-preserving local inference	Mistral Research License (non-commercial) - requires commercial license for production/clinical use
Cogito 3B	Deep Cogito	3B	Hybrid reasoning (direct + reflective modes); IDA training outperforms RLHF; strong native tool calling; 30+ languages	Clinical decision support, multi-step STEM, on-demand reasoning, agentic tool-calling	Preview checkpoint; recommends Q8 quantization (higher VRAM); newer ecosystem with fewer integrations

IBM Granite Family

Model	Dev	Params	Key Strengths	Best Workflow	Cautions
Granite 3.0 2B Dense	IBM	2B	Outperforms Llama-3.2-1B on MMLU/math/code; enterprise safety guardrails; Apache 2.0; 12T training tokens	Lightweight text generation, summarization, RAG retrieval, baseline text-only LLM	No reasoning/CoT; no vision; superseded by 3.2 and 4.x; limited math depth
Granite 3.2 2B	IBM	2B	Toggleable chain-of-thought (thinking on/off); Vision variant matches 5x larger models on DocVQA/ChartQA; 128K context	Enterprise document understanding + optional reasoning; fast Q&A; + occasional step-by-step	Reasoning mode experimental; 2B limits CoT depth; vision variant adds memory overhead
Granite 4.0 3B (H-Micro)	IBM	3B	Hybrid Mamba-2/Transformer (9:1 ratio); 70% less memory, 2x faster; linear long-context scaling (128K in ~4GB); Apache 2.0	Memory-constrained edge deployment, long-context RAG without chunking, multi-session concurrent inference	Mamba hybrid not fully supported in llama.cpp/PEFT; no CoT mode; fewer GGUF variants available
Granite 4.1 3B	IBM	3B	Best raw capability in Granite family: tool calling, coding, math, instruction following; standard dense transformer; 512K context	Production agentic workflows, function calling, API orchestration, coding assistance, domain fine-tuning	No Mamba efficiency gains; no toggleable thinking; newest model (mid-2026), less community validation

Stability AI & BigCode

Model	Dev	Params	Key Strengths	Best Workflow	Cautions
StableLM Zephyr 3B	Stability AI	3B	60% smaller than 7B yet competitive (MT-Bench 6.64); DPO-tuned for instruction following; edge-optimized	Edge chat assistants, instruction-following Q&A;, summarization where latency matters	English-only; weak math (GSM8K 42.3%); no code generation; modest MMLU (46.3)
Stable Code 3B	Stability AI	2.7B	Matches Code Llama 7B despite 60% smaller; 18 programming languages; FIM support; edge throughput measured	IDE code completion, fill-in-the-middle, lightweight coding on edge hardware	Base model not instruction-tuned (use Instruct variant for chat); surpassed by StarCoder2-3B; code-only
StarCoder2 3B	BigCode	3B	Outperforms StarCoderBase-15B (5x smaller); 600+ languages; 16K context; FIM support; fully open weights + training data	Transparent/auditable code completion for regulated environments, multi-language code generation	Base model not instruction-tuned; weak math (GSM8K 27.7); OpenRAIL license has use-case restrictions
Orca Mini 3B	Community	3B	Trained on GPT-4 explanation traces; better reasoning than typical 3B; runs in ~2GB at Q4; easy Ollama deployment	Entry-level edge chatbot, lightweight Q&A; and explanation tasks	Older OpenLLaMA architecture; known whitespace generation issues; hallucinates more than newer 3B models

LG AI, Liquid AI & Community

Model	Dev	Params	Key Strengths	Best Workflow	Cautions
EXAONE 3.5 2.4B	LG AI Research	2.4B	Highest instruction-following at its size; 32K long-context; bilingual Korean-English; 6.5T training tokens	Bilingual Korean-English workflows, long-context RAG, clinical/business translation, precise instruction compliance	Research-only license (commercial requires contacting LG); weaker in non-Korean/English languages
LFM 2.5 2.6B	Liquid AI	2.69B	Hybrid architecture (22 conv + 8 attention); agentic RL trained (BFCL 56.88); 128K context in <2.5GB; ~30 tok/s on phones	On-device AI agents with tool calling, long-context RAG on edge, data extraction pipelines, multi-step agentic workflows	Not for agentic coding or knowledge-heavy tasks; hybrid architecture less mature in tooling; limited knowledge at 2.6B
LFM 2.5 230M	Liquid AI	230M	213 tok/s on phone, 42 tok/s on Pi 5; only 293-375MB at Q4; tool use + data extraction at tiny scale; fine-tunable	Routing/skill-selection layer for robotics, data extraction pipelines, ultra-low-latency classification on CPU	Not for reasoning, math, code, or creative writing; MMLU-Pro 20.25; a fine-tuning base, not a general assistant
SmallThinker 3B	PowerInfer	3B	O1-style chain-of-thought reasoning in 3B; step-by-step traces before answering; STEM reasoning above weight class; edge-optimized	Edge reasoning tasks, math/logic decomposition, STEM education, clinical decision support with transparent reasoning	English-only; preview model (may be unstable); reasoning traces add latency; inherits Qwen2.5-3B knowledge limits

Part 3: Quick-Pick Guide - Right Model for the Job

Recommendations based on benchmark speeds, model strengths, and Jetson 8GB constraints. 'Verified' = tested on this device.

Task	Recommended Model	Why	Verified
Fastest generation	Gemma 4 E2B (gemma4:e2b)	27.1 tok/s - fastest on Jetson; 2B effective params	Yes - 27.1 tok/s
Best generalist chat	Qwen 2.5 3B Instruct	Strongest math + multilingual + good instruction following at 3B	Yes - 21.3 tok/s
Best reasoning per param	Phi-3 Mini 3.8B	69% MMLU, rivals Mixtral 8x7B; edge-optimized	Yes - 21.2 tok/s
Best function calling / agentic	Hermes 3 3B	Full fine-tune with dedicated tool-use training; JSON structured output	Yes - 20.7 tok/s
Best code completion	StarCoder2 3B	600+ languages, 16K context, FIM, outperforms 15B models	Downloading
Best code chat/assistant	Qwen 2.5 Coder 3B	Instruction-tuned code specialist, 92+ languages, same 3B footprint	Downloading
Best multimodal (vision/audio)	Gemma 3n E2B	Text+image+audio+video in ~2GB VRAM via MatFormer	Downloading
Best long context (128K+)	Llama 3.2 3B or Ministral 3B	128K-256K context; Llama has best ecosystem, Ministral has 256K	Llama: Yes 20.6 tok/s
Best memory efficiency	Granite 4.0 3B H-Micro	Mamba hybrid: 70% less memory, 2x faster, linear context scaling	Downloading
Best reasoning + tool calling	Cogito 3B	Dual-mode (direct/reflective); IDA training; native tool calling	Downloading
Best edge agent (tool use)	LFM 2.5 2.6B	Agentic RL trained; 128K in <2.5GB; hybrid architecture	Yes - 25.1 tok/s
Best clinical document understanding	Granite 3.2 2B (Vision)	DocVQA 0.89, ChartQA 0.87; toggleable reasoning; physician-friendly	Downloading
Best bilingual (Korean/English)	EXAONE 3.5 2.4B	Top instruction following + long context; Korean-English bilingual	Downloading
Smallest viable assistant	Gemma 3 1B	~529MB int4; 32K context; runs on phones and embedded	Downloading
Best STEM reasoning on edge	SmallThinker 3B	O1-style thinking traces; step-by-step decomposition at 3B	Downloading
Best ultra-light routing/classification	LFM 2.5 230M	213 tok/s on phone; 293MB; tool use at tiny scale	Yes (prior tests)
Best enterprise baseline	Granite 4.1 3B	Tool calling + coding + math; Apache 2.0; ISO certified; fine-tune ready	Downloading
Best toggleable reasoning	Granite 3.2 2B	Switch thinking on/off per query; no latency cost when off	Downloading

Part 4: License & Clinical Notes

License Type	Models	Clinical/Commercial Use
Apache 2.0	Qwen 2.5/3.5, Granite 3.0/3.2/4.0/4.1, StarCoder2, SmallThinker, LFM 2.5	Fully open - commercial and clinical use permitted
Llama 3 License	Llama 3.2 3B, Hermes 3 3B, Cogito 3B	Permitted with restrictions (>700M MAU requires Meta agreement)
MIT	Phi-3 Mini	Fully open - commercial and clinical use permitted
Mistral Research License	Ministral 3B	Non-commercial research only; commercial license required for production/clinical
OpenRAIL	StarCoder2 3B (alt view)	Use-case restrictions apply; check terms for clinical deployment
Research-only	EXAONE 3.5 2.4B	Contact LG AI Research for commercial/clinical licensing
Gemma Terms of Use	Gemma 2/3/3n, CodeGemma	Permitted with Google's acceptance of terms; redistribution allowed

Clinical note: Hermes provides drafts and support only. Walker must approve all diagnosis, prescriptions, orders, and disposition. Models marked 'research-only' or 'non-commercial' require explicit licensing before clinical deployment. Preview models (Cogito 3B, SmallThinker 3B) may have instability - validate outputs carefully.

Jetson 8GB constraint: ~4.9GB available RAM for model weights. All Q4_K_M quantized 2B-3B models fit comfortably. Q8 quantization (Cogito 3B recommendation) uses ~3.2GB weights + context buffer - still fits but with less headroom.